

Element L

Though we have a completed prototype for our project, we aren't done yet with a final solution. There are a few iterations that could be made for this project in the future. As of right now, our 'locking mechanism' is just a thick rubber layer attached to the rod. In the future, a great improvement would be a plausible mechanism which actually allows the user to lock the handle into place at different heights. To complete this, we would need to 3D print some sort of plastic piece that inserts inside the PVC pipe. We would need to create a CAD or an online shape of this beforehand though. This would take more time and money than just adding the rubber stopper. One of our group members would need to take the time to learn how to use CAD OnShape as it isn't an easy technology to get the hang of. We would give this person a couple of days to learn and design the locking mechanism. This would take away from one person's time building the handle or documenting our progress. This idea would make our design more presentable and viable, but would take much longer to complete. Another thing that we could do is make a better hooking mechanism to the cart so that the handle wiggles even less. Even though our cart handle doesn't move much now, there is still a little fluctuation when the cart is being pushed, especially with more on the cart. Creating a mechanism to attach the cart at the bottom would increase the reliability of our design. This would mean that we would need more materials which might impede on our budget of thirty dollars.

The main issue that we had while creating our prototype was not planning enough. For example, when we were trying to drill holes in the wood rods, we chose too big of a drill and the wood broke off. Even though we were able to bounce back and try to use this as a new design, more planning would have made the journey a little bit easier. Maybe in the future, we could eliminate these extra challenges by making a step by step process for each day of what everyone was going to do. Not an extensive list, but a few goals that needed to get done that day. Other than that, our group didn't have too much trouble on the parts of the Engineering Design Process.